

1st Subject

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Knowledge-Based-Systems like ...

Consider an IS-A ontology of concepts and instances. For example Tom is an instance of cat, Jerry and Speedy Gonzales are instances of mouse and Tweety is an instance of canary, Foghorn Leghorn is an instance of rooster (Gallus gallus , male chicken, with the female being called a hen), and the obvious relationships that cat, mouse, dog, ... are mammals, canary and rooster are birds, birds and mammals are animals, ... ***Design a database to store this ontology of knowledge. Find the distance between two instances.*** Distance between Jerry and Speedy Gonzales is 0, they are both instances of mouse. Distance between Tom and Jerry is 2, they are instances of cat which IS-A mammal and respectively mouse which IS-A also a mammal. Distance between Jerry and Tweety is 4, they are instances of mouse and respectively canary, mouse IS-A mammal and mammals are (IS-A) animal(s), canary IS-A bird and birds are (IS-A) animal(s). Distance between Jerry and Tweety is 3, if we consider Tweety as an instance of bird. *Can such a distance be NULL?*